

Ballistic Missile

Type of ballistic Missile

	Range
1. Short range Ballistic	under 1,000 km
2. Medium range Ballistic	1000 to 3,000 km
3. Intermediate range Ballistic	3000 to 5500 km
4. Intercontinental Ballistic	above 5500 km

short range ballistic: used or designed for fast, high-precision regional strikes against immediate cross-border.

medium range: used across wider region or neighboring countries.

Long-distance: Bridges the gap between regional warfare and continental deterrence.

Intercontinental Ballistic: Massive multi stage rocket capable of flying into sky and (space) and target across different continent.

Our neighboring country

#.	Country	Ballistic anal.
1.	Bangladesh	X
2.	China	✓
3.	Pakistan	✓
4.	Nepal	X
5.	Bhutan	X
6.	Myanmar	X
7.	Afghanistan	✓
8.	Sri Lanka	X
9.	Maldives	X

China's Ballistic Missile

Short range

DF 11 (CSS-7) : (~300 km)

DF 15 (CSS-6) : (600-900 km)

→ DF 16 (~1000 km) [newly, highly precise]

medium range

DF-21

→ DF-21D (variant) → First anti-ship ballistic missile (~2,000) km.

DF 17 (with hypersonic glide vehicle)

intermediate range

DF-26 dual capable (nuclear / conventional)
(~4,000 km)

intercontinental (ICBM)

DF-5 : (13,000 km)

DF-31 (7,000-11,000+ km range)

DF-41 (~15,000 km range)

Pakistan's Ballistic

short range:

Hatf-IX (Nasr) : (under 1000 km range)

Hatf-II (Abdali) & Hatf-III (Ghaznavi)
(200 to 300 km range)

Hatf-IV (Shaheen-I/IA) : (750 to 900 km)

medium range:

Hatf-V (Ghauri) : (North-Korean tech.)
range : (2000 - 5000 km)

Hatf-VI (Shaheen-II) (~2000 - 2500 km)

Shaheen-III : (Pakistan's longest range missile)
range : (app. 2750 km).

Ababeel : (MIRV technology)

Afghanistan itself doesn't possess any ballistic missile but contains past war leftover weapons.

The two of their so-called existing ballistic (left over war) are:

- a) Scud-B (300 km)
- b) FROG-7 (70 km)

Ballistic Kinetics

Basically, Ballistic follows the projectile Motion

Factor affecting:

- 1. If speed change $\longleftrightarrow$ height change
- 2. If height change $\longleftrightarrow$ distance change
- 3. Both speed and height change $\longleftrightarrow$ distance change

How Ballistic Missiles Work

A quick overview of principles, flight phases, and classification

What Makes It “Ballistic”

Unlike a cruise missile, which is powered and actively guided for its entire flight, a ballistic missile is only powered during the early part of its journey. Once the engines cut off, it follows a curved, gravity-driven path — much like a thrown ball, but at far greater speed and altitude. This unpowered arc is what gives the missile its name.

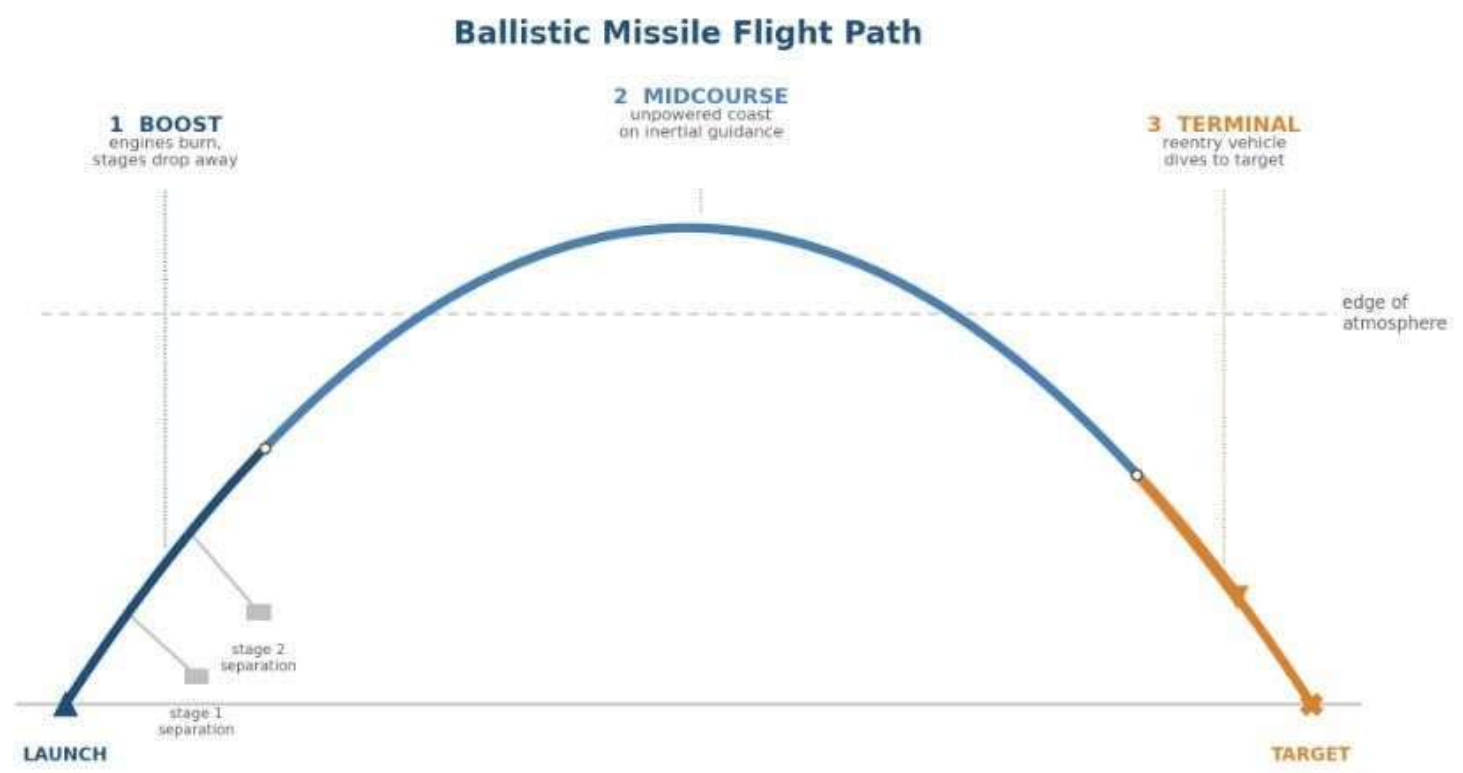
The Three Flight Phases

1. Boost Phase Rocket engines (solid or liquid fuel, often in multiple stages that burn and separate) accelerate the missile upward and outward. This is the only powered portion of the flight, typically lasting just a few minutes.	2. Midcourse Phase Engines shut off and the missile coasts unpowered through the upper atmosphere or space along its ballistic arc, guided by pre-set inertial navigation, sometimes refined with satellite positioning data.	3. Terminal Phase The reentry vehicle carrying the warhead separates and dives back into the atmosphere toward the target at extreme speed, shielded by a heat-resistant covering.
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Core Components

- **Propulsion:** Multi-stage propulsion — rocket stages that burn in sequence and drop away once spent, reducing weight and increasing range.
- **Guidance:** Guidance and navigation — primarily inertial systems using gyroscopes and accelerometers to track position from the moment of launch.
- **Payload:** Payload — a conventional or nuclear warhead housed inside the reentry vehicle.

Flight Path Diagram



Simplified trajectory showing the three flight phases and stage separation.